MyDentist End-to-End BI Solution

Overview

A **production-grade Microsoft Fabric BI system** built from scratch to demonstrate modern data architecture, advanced analytics, and enterprise reporting. Designed as an interview showcase project; delivered a fully functional, interview-quality solution within scope constraints.

Architecture & Approach

Medallion Architecture

Implemented a **three-layer medallion pattern** in Fabric for scalable, maintainable data pipelines:

- **Land Layer**: Raw data ingestion from five heterogeneous source systems; minimal transformation; trace audit columns
- **Cleanse Layer**: Data quality enforcement; standardization of naming/formatting; business rule application; reconciliation checkpoints
- **Conform Layer**: Dimensional modelling; aggregations; business-ready curated datasets

This separation enables:

- Clear separation of concerns and independent layer optimization
- Rapid iteration on business logic without re-ingesting raw data
- Audit trails and quality gates at each stage

Kimball Dimensional Model

Built a **star schema** with:

- **2 Fact Tables**: Transaction-level and summary-level (conformed dimensions across both)
- **6 Dimension Tables**: Patient, Provider, Service, Location, Time, and Reference dimensions
- **Conformed Dimensions**: Shared dimension keys across fact tables ensure consistency and enable drill-through analysis
- **Surrogate Keys**: Type 2 slowly-changing dimensions for provider and patient dimension tracking historical changes

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Technical Stack & Techniques

Data Integration & Transformation

- **Microsoft Fabric** (Synapse Spark notebooks for ELT)

- **PySpark** for scalable data cleaning and transformation
- **Incremental loading** patterns with watermarking to minimize reprocessing
- **Delta Lake** for ACID compliance and time-travel queries

Data Modelling

- **Kimball methodology** with conformed dimensions and fact table linearity
- **Factless fact tables** for dimensional analysis (e.g., patient-provider relationships)
- **Junk dimensions** for low-cardinality attributes to reduce dimension table bloat

Analytics & Visualization

- **Tabular Editor** for enterprise DAX development (40+ measures)
- **Advanced DAX patterns**:
  - Time intelligence (`PARALLELPERIOD`, `DATESBETWEEN`)
  - Complex aggregations with context transitions (`CALCULATE`, `ALL`)
  - Ratio calculations (e.g., revenue YoY growth, utilization %)
  - Dynamic filtering and slicers
- **Power BI** (three-page interactive report)
  - Drill-through capabilities between report pages
  - Tooltips with underlying detail
  - Performance optimization via aggregation tables and field parameters

Source Systems

Integrated data from **5 distinct sources**:

1. Patient management system (appointments, demographics)
2. Financial/billing system (invoicing, payments)
3. Clinical records (treatment codes, outcomes)
4. Provider scheduling (availability, capacity)
5. Reference tables (fee schedules, service catalogue)

Data Model Summary

Fact Tables

- **Appointment Fact**: Granularity = one row per appointment (date, provider, patient, service, revenue, cost)
- **Clinical Outcome Fact**: Treatment-level detail (procedure codes, success indicators, follow-up flags)

Dimension Tables

1. **Patient Dim**: Patient ID, demographics, registration date, active status (Type 2)
2. **Provider Dim**: Provider ID, credentials, specialty, location assignment (Type 2)
3. **Service Dim**: Service code, description, category, base fee
4. **Location Dim**: Clinic ID, address, region, capacity
5. **Time Dim**: Date, fiscal period, day-of-week, holiday flags
6. **Reference Dim**: Lookup values (appointment status, outcome codes, payment methods)

Key DAX Measures

Examples of 40+ production measures built in Tabular Editor:

```
Total Revenue = SUM(AppointmentFact[Revenue])

YoY Growth % = DIVIDE(
    [Current Year Revenue] - [Prior Year Revenue],
    [Prior Year Revenue],
    0
)

Provider Utilization % = DIVIDE(
    COUNTROWS(AppointmentFact),
    [Scheduled Capacity],
    0
)

Patient Retention (30D) = CALCULATE(
    DISTINCTCOUNT(PatientDim[PatientID]),
    FILTER(AppointmentFact,
        AppointmentFact[AppointmentDate] >= TODAY()-30
    )
)
```

Report Outputs

Three-page interactive Power BI report:

1. **KPI Dashboard**: Summary metrics (revenue, appointment count, utilization, patient retention)
2. **Provider Performance**: Drill-able provider scorecard (revenue, appointment volume, patient satisfaction)

3. **Patient Analysis:** Cohort behaviour, repeat visit patterns, service uptake trends

Features:

- Cross-page drill-through from KPIs to detail pages
- Date slicers with fiscal calendar support
- Location and specialty filters
- Tooltip detail views showing underlying transactions

Outcomes & Impact

Interview Performance

- **Demonstrated** end-to-end BI capability from source systems through production reporting
- **Impressed** technical panel (James, hiring manager) with architectural decisions and clean code
- **Validated** ability to own a project independently: requirements → design → delivery

Technical Achievements

- **Scalable foundation**: Medallion architecture supports petabyte-scale extensions
- **Enterprise standards**: Tabular Editor, version control, DAX naming conventions
- **Business value**: Reporting suite immediately actionable by stakeholders

Decision & Outcome

- Declined £60k 12-month FTC offer from mydentist (January 2026)
- Reason: Signed contract with **Footasylum** (Senior BI Developer, £60k, permanent role, starting 1 July 2026)
- Project served its purpose: credible proof of full-stack BI delivery in an interview context

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Key Learnings

1. **Scope discipline**: Built a production-quality model within interview constraints (time-boxed delivery)
2. **Architecture-first thinking:** Medallion pattern decisions made project maintainability and extension trivial
3. **Stakeholder readiness**: Three-page report proved sufficient for interview narrative; didn't over-engineer
4. **Tabular Editor mastery**: Professional DAX development using IDE-grade tooling vs. Power BI GUI

Contact

Built by: Kash | Senior BI Developer
Date: January 2026
Stack: Microsoft Fabric, Spark, PySpark, DAX, Power BI, Tabular Editor

For questions on the approach or architecture: replicable patterns; transferable to any analytical domain.